

Digital Material Passport

ID DMP-2024-ARR-001 - Version 0.1.0
Issue Date 2024-12-15 - Certificate Type EN 10204 3.1

Manufacturer	Customer	Business Transaction
Advanced Steel Technologies GmbH Industriestraße 15 52070Aachen DE quality@advanced-steel.de	Precision Engineering Ltd. Manufacturing Park 42 Birmingham B1 1AA GB procurement@precision-eng.co.uk	Order ID PO-2024-7890 Delivery ID DN-2024-3456 - Quantity 500 kg

Product

Product Name	Hardenability Test Steel	Shape	RoundBar - D 100 mm - Length 1000 mm
Batch ID	HTB-2024-045		

Chemical Analysis

Heat Number H-2024-078 - Sample Location Ladle

Symbol	Unit	Min	Max	Actual	Symbol	Unit	Min	Max	Actual
C	%	0.42	0.5	0.45	S	%		0.025	0.008
Mn	%	0.5	0.8	0.65	Cr	%		0.25	0.12
Si	%		0.4	0.25	Ni	%		0.25	0.08
P	%		0.03	0.015					

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Jominy Hardenability Test					ISO 377	✓

Quenched from 850°C

Distance from - quenched end (mm)	1.5	3	5	7	9	11	15	20	25	30
Value [HRC]	45	43	40	36	33	31	27	25	28	27
Min	42	39	35	32	29	26	22	20		
Max	47	46	44	41	39	37	33	31	<= 30	<= 29
Status	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass

Coating Thickness Profile

Magnetic induction method

ISO 2178 ✓

Position along length (mm)	0	50	100
Value [µm]	125	115	108
Min	>= 120	>= 110	>= 100
Max	<= 130	<= 125	<= 120
Status	Pass	Pass	Pass

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with EN 10083-1 and ISO 377. All hardenability requirements have been met.

Validated by **Dr. Maria Schmidt**, Metallurgical Engineer,
Quality Control Laboratory - 2024-12-15